TEAM

SkyMate

AGENTIC FLIGHT ASSISTANT · DEL → BOM → ANYWHERE

MEMBER 01
Member name

MEMBER 02
Member name

MEMBER 03
Member name

MEMBER 04
Member name

MEMBER 05
Member name

MENTOR Mentor name

Click any name to type it in. Your names are kept in this browser.

PROBLEM

Planning one flight still takes five tabs and a leap of faith

01 **Scattered information.** Schedules, airline rules, terminals, weather and documents live in different places.

02 **Untrustworthy answers.** Generic chatbots invent fares, gates and timings with no fact-check.

03 **No direct flight, no answer.** Finding connections takes many searches and costs API calls.

04 **No human control.** Costly or irreversible steps, like booking, run without the traveller's go-ahead.

05 **Locked to one use case.** Moving to a new domain means rebuilding the assistant from scratch.

SOLUTION

One chat that plans, checks and asks before it acts

Full walkthrough in the live demo

CHAT **Understands plain requests** and searches live flights worldwide, including one-stop connections.

AGENTS **A 12-agent team led by a supervisor** builds the itinerary: 9 specialists in parallel, a web agent fills gaps.

SAFETY **Guardrails and fact-checks** block injection attempts and check every answer against real data.

HUMAN **Clearances queue** for approving connection searches, itinerary reviews and bookings, or saving drafts.

TRIP **Your documents, live weather, SkyPay checkout,** calendar and PDF in one place.

GENERI **Domain packs:** change one folder to reuse the whole engine for a new use case.

TECH STACK

Built on LangGraph, LangChain and live data

12	16	3	33	1
agents per itinerary	orchestrated graph steps	human checkpoints	REST endpoints	folder per domain

AI models

via the company LLM gateway

GPT-5-mini	itineraries
GPT-4.1-mini	chat · agents
text-embedding-3-large	RAG
Whisper	voice notes

Orchestration

two supervised graphs

LangGraph	0.4.3
StateGraph · Send	parallel agents
interrupt · Command	pause / resume
SQLite checkpointer	2.0.10

LangChain

model calls and documents

langchain-core	1.6.9
langchain-openai	1.7.0
text-splitters	RAG chunks

Safety & control

trust built in

Rule guardrails	SQL · code · jailbreak
LLM safety classifier	every message
Grounding fact-check	replies · plans
Clearances queue	human approval

Data & memory

stored locally

ChromaDB	1.0.9 · vectors
SQLite	approvals · runs
JSON + bcrypt	users

Live data

real-world facts

AeroDataBox	schedules
Open-Meteo	weather
Tavily → Wikipedia	web search

Backend

Python 3.12

Flask	3.1.3
Flasgger	Swagger docs
JWT · PyJWT	auth
fpdf2	trip PDF

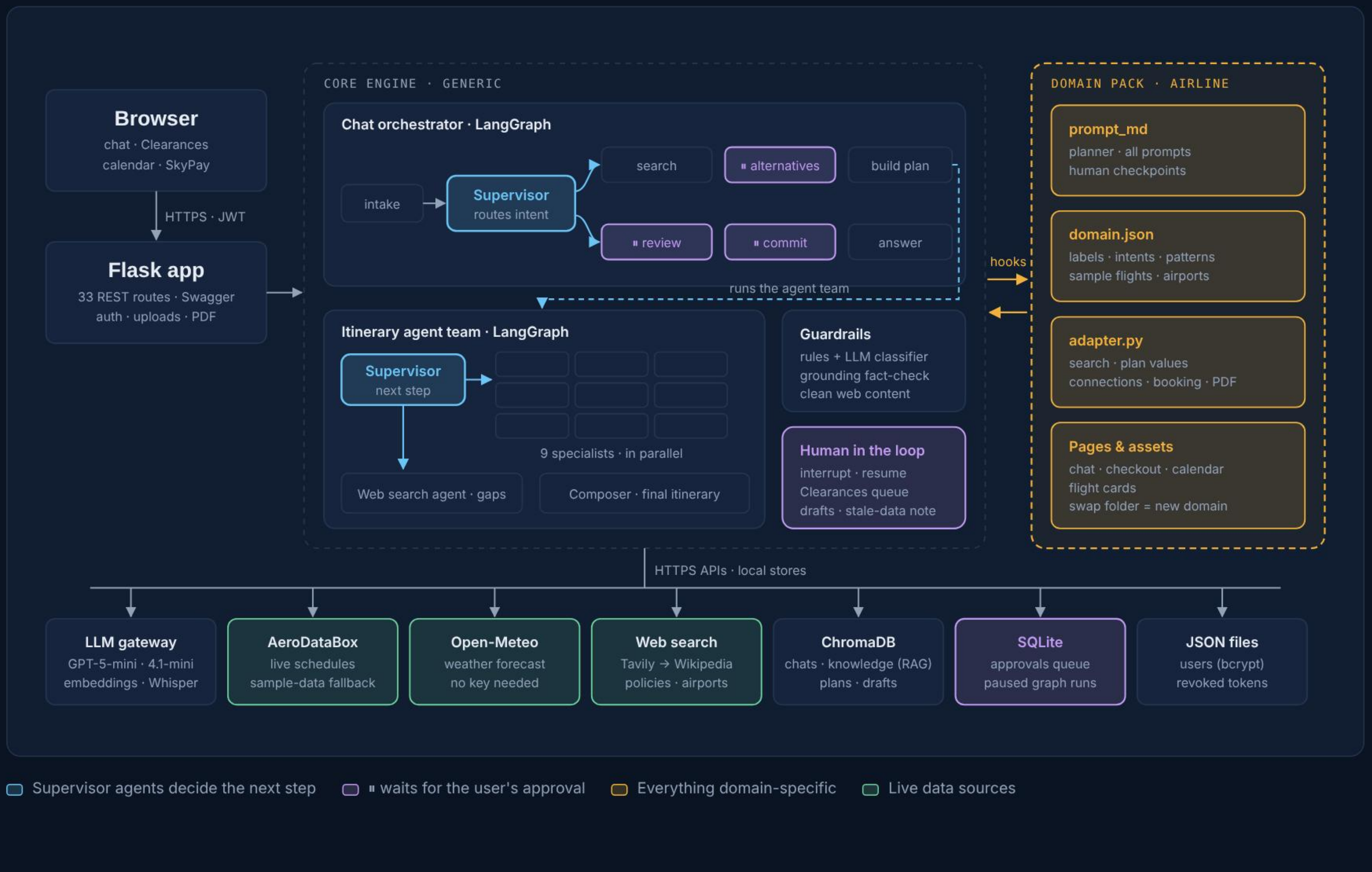
Frontend

server-rendered

Jinja templates	domain-aware
Vanilla JS	no framework
SkyPay checkout	demo payments

ARCHITECTURE

A generic engine with a plug-in domain pack



Supervisor agents decide the next step · ⏸ waits for the user's approval · Everything domain-specific · Live data sources